

ABDUL ZACKY

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PROFESSIONAL SUMMARY

Results-driven Computer Science student at Universitas Indonesia with strong expertise in Machine Learning, Deep Learning, and Data Science. Proven track record of developing AI-powered solutions with measurable impact, including projects that achieved 99% accuracy in document verification and projected Rp 192.5B in annual savings. Experienced in PyTorch, TensorFlow, scikit-learn, and advanced NLP techniques. Winner of multiple national AI/ML hackathons and passionate about leveraging data science to solve real-world problems.

EDUCATION

Universitas Indonesia

Bachelor of Computer Science | GPA: 3.71/4.00 | Expected Graduation: 2027

Relevant Coursework: Machine Learning, Deep Learning, Database Systems, Linear Algebra, Calculus, Data Structures & Algorithms

TECHNICAL SKILLS

Programming Languages: Python, Java, SQL, Dart

Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, Computer Vision, NLP, Generative AI

Data Science & Analytics: Pandas, NumPy, Data Preprocessing, Feature Engineering, Model Evaluation, Statistical Analysis

NLP & Text Processing: spaCy, NLTK, Named Entity Recognition, Text Classification, Sentiment Analysis, OCR

Databases & Tools: SQL, PostgreSQL, MySQL, Git, Jupyter Notebook

Web Development: Flutter, React, Next.js, Full-stack Development

Soft Skills: Problem Solving, Team Collaboration, Communication, Teaching, Project Management

PROFESSIONAL EXPERIENCE

IT Developer & Data Analytics Intern | Telkom Indonesia

July 2025 – August 2025 | Jakarta, Indonesia

- Developed data-driven insights and dashboards using SQL and data analytics tools to support strategic decision-making
- Performed data analysis and visualization to identify trends and patterns in large datasets
- Collaborated with cross-functional teams to implement business intelligence solutions

Scientific Committee - Data Analytics & Dashboard | COMPFEST Fasilkom UI

February 2025 – November 2025 | Depok, Indonesia

- Contributed to the development of data-driven insights and dashboards for Indonesia's largest student-run IT event
- Ensured accurate analysis, visualization, and strategic decision-making for COMPFEST's scientific initiatives
- Implemented data analytics solutions to track event metrics and participant engagement

Teaching Assistant - Databases (SQL) | Universitas Indonesia

January 2025 – June 2025 | Depok, Indonesia

- Facilitated learning in database management systems by teaching SQL concepts to 50+ students
- Assisted students with assignments and projects involving complex SQL queries and database design
- Fostered deeper understanding of relational database management systems and normalization techniques

Teaching Assistant - Linear Algebra & Calculus | Universitas Indonesia

July 2024 – December 2024 | Depok, Indonesia

- Guided students in mastering Linear Algebra and Calculus concepts essential for machine learning
- Provided clear explanations and personalized guidance to enhance students' academic performance
- Developed supplementary materials and practice problems to reinforce mathematical foundations

Staff of Business Development | COMPFEST Fasilkom UI

April 2024 – February 2025 | Depok, Indonesia

- Drove growth and success by identifying business opportunities and building strategic partnerships
- Contributed to securing sponsorships and partnerships worth millions of rupiah for the event

KEY PROJECTS & ACHIEVEMENTS

SIVANA - AI-Powered Civil Servant Recruitment System

Technologies: Computer Vision, Generative AI, OCR, NLP, Face Verification, Liveness Detection | sivana.id

- Developed end-to-end AI solution revolutionizing Indonesia's CASN selection process with 99% accuracy (from 80%)
- Implemented automated document verification using OCR and NLP, reducing verification time from 2-3 weeks to <1 minute
- Built cheat-proof face verification system with liveness detection, projecting 90% reduction in fraud cases
- Created auto-generated TMS feedback system, projecting Rp 192.5B in annual savings
- Won 1st Place at ASN Digital AI Hackathon 2025

Quikyu - Smart Queue System for PT KAI

Technologies: Machine Learning, AI Demand Prediction, Behavior Analysis, Bot Detection | quikyu.xyz

- Built multi-layer ML solution to enhance security and user experience for PT KAI's digital booking platform
- Implemented AI-powered demand prediction to proactively manage high traffic scenarios
- Developed behavior-based anti-bot system to prevent fraudulent activity and ensure fair ticket distribution
- Created non-intrusive smart waiting room system to optimize user flow during peak demand
- Won 1st Place at Hacksphere, President University Compsphere by KAI

Petatas - Financial Literacy AI Platform for Papua

Technologies: Generative AI, Gamification, Educational Technology | petatas.vercel.app

- Created innovative Generative AI-based solution to bridge financial literacy gap in Papua
- Addressed centralization of QRIS transactions by gamifying the learning process
- Designed educational ecosystem to increase QRIS usage outside of Sorong City
- Won 1st Place at Digital Innovation Challenge Bank Indonesia Papua Barat

DataClair - Automated Machine Learning Platform

Technologies: Random Forest, XGBoost, Automated ML, Feature Engineering, Python | dataclair.vercel.app

- Built full-stack web application generating ML predictions from CSV data without coding requirements
- Implemented automated preprocessing, feature engineering, and intelligent model selection
- Enabled real-time predictions with downloadable results and performance feedback system
- Deployed scalable solution handling multiple concurrent users and large datasets

Neural Style Transfer Implementation

Technologies: PyTorch, Deep Learning, Computer Vision, Convolutional Neural Networks

- Implemented neural style transfer using deep learning to blend artistic styles with content images
- Extracted style features at multiple levels and applied them while preserving structural integrity
- Optimized model performance and training efficiency using PyTorch framework

Sinavika - BPJS Kesehatan Enhancement System

Technologies: AI, INA-CBG Integration, Healthcare Analytics | sinavika.vercel.app

- Enhanced BPJS Kesehatan system using AI e-klaim analysis with INA-CBG integration
- Implemented IGD analysis and personal healthcare AI with end-to-end integration
- Connected hospital, patient, and BPJS systems for seamless data flow and improved healthcare delivery

AWARDS & HONORS

- 1st Winner - ASN Digital AI Hackathon 2025 (SIVANA - AI-powered civil servant recruitment)
- 1st Winner - Hacksphere, President University Compsphere by KAI (Quikyu - Smart Queue System)
- 1st Winner - Digital Innovation Challenge Bank Indonesia Papua Barat (Petatas - Financial Literacy AI)
- 2nd Winner - NexSolve Research Paper Competition, TECHTONIC UI 2025 (AI-Powered Passenger Distribution)